

LANZHOU, China

WMO: 528890

Lat: 36.050N

Lon: 103.883E

Elev: 4980

StdP: 12.24

Time Zone: 8.00 (E08)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	14.3	-3.8	5.4	24.6	-1.2	6.2	23.9	7.5	28.6	6.7	28.9	1.1	90	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	90.7	64.5	87.9	63.3	85.4	62.3	68.5	83.4	67.1	81.5	65.7	79.7	5.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	107.9	76.2	62.5	101.8	74.3	60.9	96.2	72.7	36.1	83.3	34.8	81.3	33.5	80.0	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	7.8	7.0	7.2	96.3	3.7	3.4	4.5	98.7	2.3	100.7	0.2	102.6	-2.5	105.0		
			DB	WB	5.5	71.1	3.3	1.4	3.1	72.1	1.2	72.9	-0.7	73.7	-3.1	74.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	25.7	34.1	44.5	55.5	63.8	70.6	74.3	71.9	64.2	51.9	39.5	28.2
	DBStd	17.64	4.58	5.70	6.89	7.58	6.22	4.76	5.40	5.11	5.40	5.61	5.92	4.69
	HDD50	2467	753	445	196	31	3	0	0	0	0	47	316	676
	HDD65	5516	1218	865	636	296	94	11	2	6	76	406	765	1141
	CDD50	3238	0	0	25	196	431	619	753	679	428	106	1	0
	CDD65	811	0	0	0	11	57	180	290	219	54	0	0	0
	CDH74	6879	0	0	3	149	553	1499	2591	1714	367	3	0	0
	CDH80	2301	0	0	0	36	114	453	1051	563	84	0	0	0
Wind	WSAvg	1.9	0.9	1.6	2.4	2.7	2.8	2.8	2.7	2.3	1.7	1.3	0.9	0.8
Precipitation	PrecAvg	12.5	0.0	0.1	0.3	0.7	1.5	1.6	2.5	2.8	1.9	0.9	0.2	0.0
	PrecMax	21.1	0.3	0.4	0.9	2.7	5.0	4.3	5.6	6.1	3.6	2.4	0.8	0.2
	PrecMin	7.4	0.0	0.0	0.0	0.0	0.0	0.5	0.6	0.3	0.4	0.0	0.0	0.0
	PrecStd	3.0	0.1	0.1	0.2	0.6	1.2	0.9	1.3	1.5	0.9	0.7	0.2	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	56.7	71.6	85.3	86.9	91.5	96.8	92.8	87.7	73.1	60.9	44.6
		MCWB	32.6	40.0	49.1	55.6	56.8	60.2	67.2	66.0	60.6	54.4	45.9	33.7
	2%	DB	40.8	52.1	65.4	78.1	83.2	87.8	91.8	88.8	81.8	69.2	57.5	41.6
		MCWB	29.9	38.1	45.8	52.3	56.3	61.4	65.7	65.3	61.2	52.7	43.8	31.9
	5%	DB	37.5	48.1	61.4	73.5	80.0	85.1	88.9	85.8	77.9	65.7	53.8	39.2
		MCWB	28.2	35.8	43.8	50.3	55.3	60.0	64.9	64.5	59.8	51.3	41.8	30.2
	10%	DB	34.5	44.7	57.2	69.2	76.8	82.3	86.1	83.0	74.7	62.4	50.2	36.9
		MCWB	26.4	33.6	41.9	49.0	54.3	59.3	64.0	63.7	59.0	49.7	40.1	28.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	41.3	50.2	57.9	61.1	66.3	71.1	70.4	68.7	57.3	47.4	34.9
		MCDB	44.3	54.1	68.1	74.6	76.3	81.0	90.5	85.6	75.7	67.9	59.2	43.5
	2%	WB	30.3	38.6	47.1	54.3	59.2	64.2	68.9	68.3	63.9	54.6	44.7	32.1
		MCDB	39.6	50.5	61.4	72.4	75.7	78.7	86.0	81.9	76.0	65.8	55.1	40.8
	5%	WB	28.4	36.3	45.0	52.1	57.6	63.0	67.4	67.1	62.0	52.9	42.6	30.5
		MCDB	37.0	46.9	58.4	70.0	73.9	78.1	83.2	80.4	73.8	62.9	52.7	38.8
	10%	WB	26.6	34.2	42.9	50.2	56.1	61.7	66.1	65.6	60.2	51.1	40.4	28.8
		MCDB	34.0	43.9	56.4	66.5	72.7	76.7	81.4	78.7	72.2	60.1	49.3	36.3
Mean Daily Temperature Range	5% DB	MDBR	18.3	19.3	19.4	20.7	20.2	19.8	19.0	18.7	18.2	18.4	18.7	17.5
		MCDBR	22.7	25.0	27.3	28.3	26.7	25.4	23.9	23.3	24.4	23.5	24.1	19.9
		MCWBR	15.1	14.6	13.3	11.1	9.3	7.6	6.5	7.4	8.7	10.6	13.5	13.2
	5% WB	MCDBR	22.1	23.4	24.6	23.9	21.9	20.1	20.0	19.3	19.2	21.2	22.7	19.6
		MCWBR	14.8	13.8	12.3	10.4	8.7	7.3	6.6	7.4	8.0	10.2	13.0	13.1
Clear-Sky Solar Irradiance	taub	0.369	0.432	0.499	0.538	0.534	0.520	0.501	0.514	0.489	0.452	0.384	0.352	
	taud	2.044	1.893	1.765	1.719	1.800	1.894	2.017	1.956	1.967	1.977	2.070	2.119	
	Ebn at Noon	251	247	241	240	244	247	251	244	242	238	243	249	
	Edh at Noon	41	53	66	73	69	62	55	57	54	49	39	35	
All-Sky Solar Radiation	RadAvg	928	1177	1498	1766	1889	1865	1818	1670	1286	1169	994	889	
	RadStd	88	83	97	103	112	107	167	119	96	108	75	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page